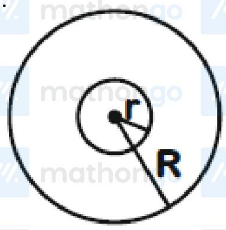
Q1 JEE Main 2020 - 2 September (Evening)

A charge Q is distributed over two concentric conducting thin spherical shells radii r and $\mathrm{R}(\mathrm{R}>\mathrm{r})$. If the surface charge densities on the two shells are equal, the electric potential at the common centre is :



(A) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)}{(R^2+r^2)}Q$
(B) $\frac{1}{4\pi\varepsilon_0}\frac{(R+2r)Q}{2(R^2+r^2)}$
(C) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)}{2(R^2+r^2)}Q$
(D) $\frac{1}{4\pi\varepsilon_0}\frac{(2R+r)}{(R^2+r^2)}Q$

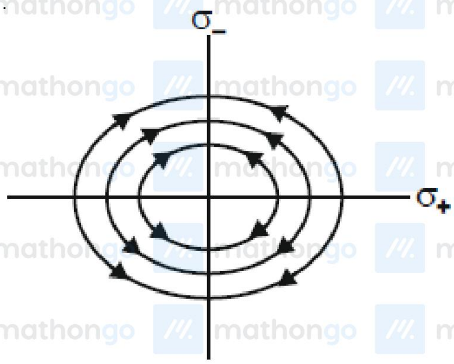
Q2 JEE Main 2020 - 3 September (Evening)

Concentric metallic hollow spheres of radii R and $4\mathrm{R}$ hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference $V(R)-V(4R)$ is

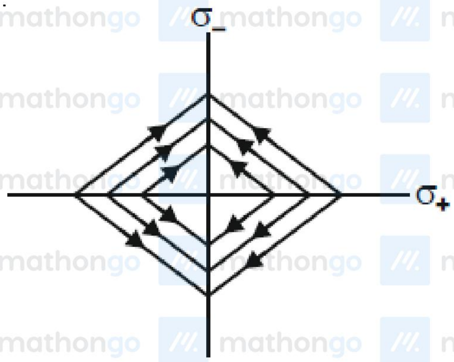
(A) $\frac{3Q_2}{4\pi\varepsilon_0 R}$
(B) $\frac{3Q_1}{16\pi\varepsilon_0 R}$
(C) $\frac{Q_2}{4\pi\varepsilon_0 R}$
(D) $\frac{3Q_1}{4\pi\varepsilon_0 R}$

Q3 JEE Main 2020 - 4 September (Morning)

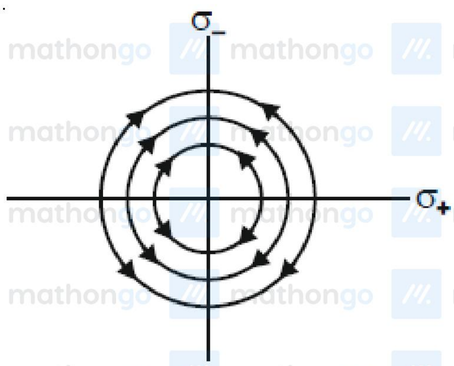
Two charged thin infinite plane sheets of uniform surface charge density σ_+ and σ_- where $|\sigma_+|>|\sigma_-|$, intersect at right angle. Which of the following best represents the electric field lines for this system?



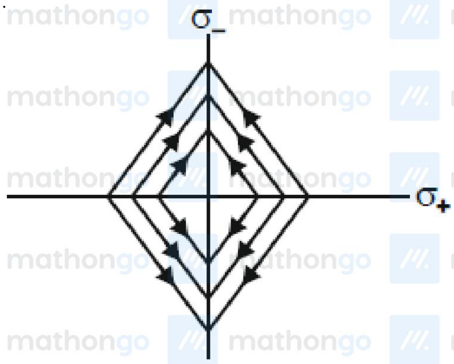
(A)



(B)



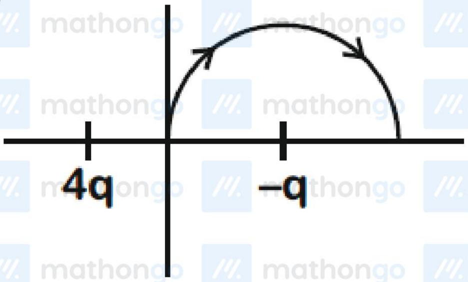
(C)



(D)

Q4 JEE Main 2020 - 4 September (Morning)

A two point charges $4\mathrm{q}$ and $-\mathrm{q}$ are fixed on the x axis at $x=-\frac{d}{2}$ and $x=\frac{d}{2}$, respectively. If a third point charge 'q' is taken from the origin to $x=d$ along the semicircle as shown in the figure, the energy of the charge will



(A) Decrease by $\frac{\mathrm{q}^2}{4\pi\varepsilon_0\mathrm{d}}$
(B) Decrease by $\frac{4q^2}{3\pi\varepsilon_0 d}$
(C) Increase by $\frac{2q^2}{3\pi\varepsilon_0 d}$
(D) Increase by $\frac{3q^2}{4\pi\varepsilon_0 d}$

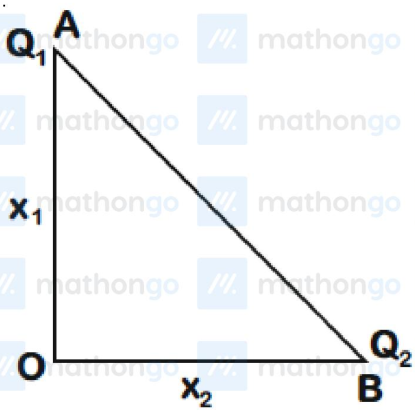
Q5 JEE Main 2020 - 5 September (Evening)

Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1,3,5,7,9 have charge $(+q)$ each, while 2,4,6,8,10 have charge $(-q)$ each.

The potential V and the electric field E at the centre of the circle are respectively.
(Take $\mathrm{V}=0$ at infinity)
(A) $\mathrm{V}=0 ; \mathrm{E}=0$
(B) $\mathrm{V}=\frac{10 \mathrm{q}}{4 \pi \varepsilon_0 \mathrm{R}} ; \mathrm{E}=\frac{10 \mathrm{q}}{4 \pi \varepsilon_0 \mathrm{R}^2}$
(C) $\mathrm{V}=\frac{10 \mathrm{q}}{4 \pi \varepsilon_0 \mathrm{R}} ; \mathrm{E}=0$
(D) $\mathrm{V}=0 ; \mathrm{E}=\frac{10 \mathrm{q}}{4 \pi \varepsilon_0 \mathrm{R}^2}$

Q6 JEE Main 2020 - 6 September (Morning)

Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1 / Q_2 is proportional to



(A) $\frac{x_1^3}{x_2^3}$
(B) $\frac{x_2^2}{x_1^2}$
(C) $\frac{x_1}{x_2}$
(D) $\frac{x_2}{x_1}$

Q7 JEE Main 2020 - 6 September (Evening)

Two identical electric point dipoles have dipole moments $\overrightarrow{\mathrm{p}_1}=\mathrm{p} \hat{\mathrm{i}}$ and $\overrightarrow{\mathrm{p}_2}=-\mathrm{p} \hat{\mathrm{i}}$ and are held on the x axis at distance 'a' from each other. When released, they move along the x-axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is

(A) $\frac{\mathbf{p}}{\mathbf{a}} \sqrt{\frac{1}{2 \pi \varepsilon_0 \mathbf{m a}}}$
(B) $\frac{\mathbf{p}}{\mathbf{a}} \sqrt{\frac{2}{\pi \varepsilon_0 \mathrm{ma}}}$
(C) $\frac{\mathbf{p}}{\mathbf{a}} \sqrt{\frac{1}{\pi \varepsilon_0 \mathbf{m a}}}$
(D) $\frac{p}{a} \sqrt{\frac{3}{2 \pi \varepsilon_0 m a}}$

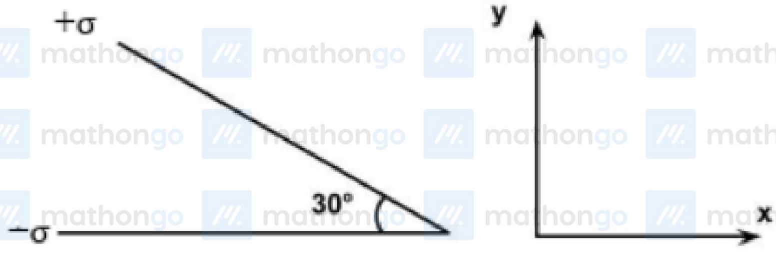
Q8 JEE Main 2020 - 6 September (Evening)

Consider the force F on a charge 'q' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F, if q ' is placed at distance r from the centre of the shell ?

(A) $\mathrm{F}=\frac{1}{4 \pi \varepsilon_0} \frac{\mathrm{Qq}}{\mathrm{r}^2}$ for all r
(B) $\frac{1}{4 \pi \varepsilon_0} \frac{\mathrm{qQ}}{\mathrm{R}^2}>\mathrm{F}>0$ for $\mathrm{r}<\mathrm{R}$
(C) $\mathrm{F}=\frac{1}{4 \pi \varepsilon_0} \frac{\mathrm{Qq}}{\mathrm{r}^2}$ for $\mathrm{r}>\mathrm{R}$
(D) $F=\frac{1}{4 \pi \varepsilon_0} \frac{Q q}{R^2}$ for $r<R$

Q9 JEE Main 2020 - 7 January (Morning)

Two infinite planes each with uniform surface charge density $+\sigma$ are kept in such a way that the angle between them is 30^o . The electric field in the region shown between them is given by :



(A) $\frac{\sigma}{2\varepsilon_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{1}{2}\widehat{x}\right]$

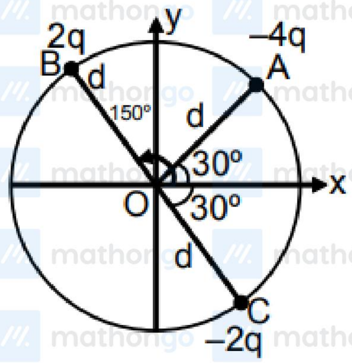
(B) $\frac{\sigma}{2\varepsilon_0}\left[\left(1+\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{1}{2}\widehat{x}\right]$

(C) $\frac{\sigma}{2\varepsilon_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}+\frac{1}{2}\widehat{x}\right]$

(D) $\frac{\sigma}{2\varepsilon_0}\left[\left(1+\frac{\sqrt{3}}{2}\right)\hat{y}+\frac{1}{2}\widehat{x}\right]$

Q10 JEE Main 2020 - 8 January (Morning)

Three charged particles A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d. The charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is:



(A) $\frac{q}{4\pi\varepsilon_0 d^2}$

(B) $\frac{q\sqrt{3}}{4\pi\varepsilon_0 d^2}$

(C) $\frac{q\sqrt{3}}{\pi\varepsilon_0 d^2}$

(D) $\frac{q\sqrt{3}}{2\pi\varepsilon_0 d^2}$

Q11 JEE Main 2020 - 8 January (Morning)

In finding the electric field using gauss law the formula $|\vec{E}|=\frac{q_{inc}}{\varepsilon_o|A|}$ is applicable. In the formula ε_o is permittivity of free space, A is the area of Gaussian surface and q_{inc} is charge enclosed by the Gaussian surface. This equation can be used in which of the following situation?

(A) Surface is equipotential

(B) Magnitude of electric field is constant
(C) Magnitude of electric field is constant and the surface is equipotential
(D) For all Gaussian surfaces.

Q12 JEE Main 2020 - 8 January (Evening)

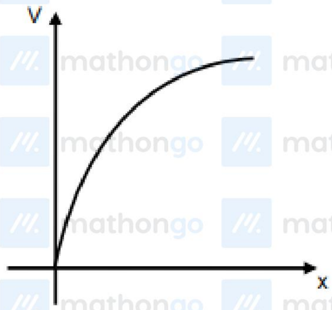
Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2, respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that E_1/E_2 = R_1/R_2. Then the ratio V_1(on S_1)/ V_2 (on S_2) of the electrostatic potentials on each sphere is

(A) $\frac{r_1}{r_2}$
(B) $\left(\frac{r_1}{r_2}\right)^2$
(C) $\frac{r_2}{r_1}$
(D) $\left(\frac{r_2}{r_1}\right)^2$

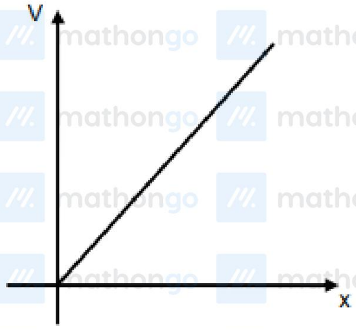
Q13 JEE Main 2020 - 8 January (Evening)

A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)

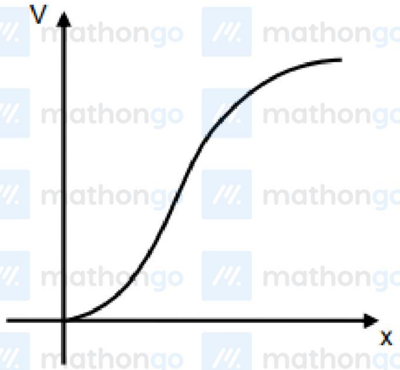
(A)



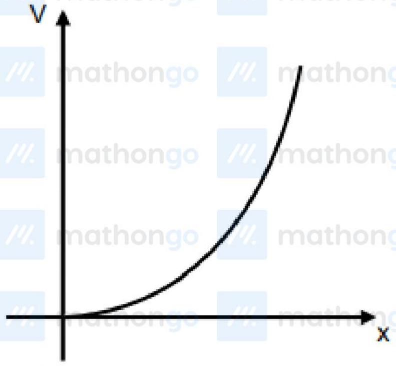
(B)



(C)

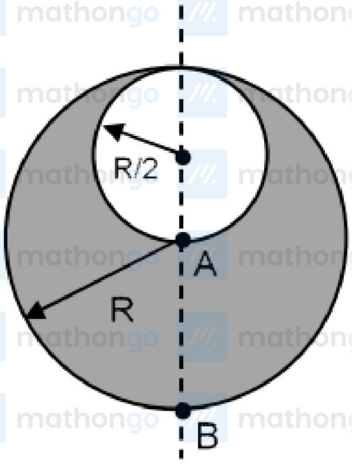


(D)



Q14 JEE Main 2020 - 9 January (Morning)

A solid sphere having radius R and Uniform charge density ρ has a cavity of radius $R/2$ as shown in figure. Find the ratio of magnitude of electric field at point A and B i.e. $\left|\frac{E_A}{E_B}\right|$.



(A) $\frac{18}{19}$
(B) $\frac{11}{17}$
(C) $\frac{9}{17}$
(D) $\frac{9}{19}$

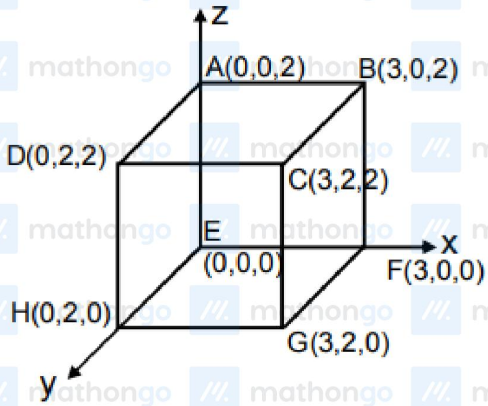
Q15 JEE Main 2020 - 9 January (Morning)

An electric dipole of moment is at the origin $(0,0,0)$. The electric field due to this dipole at $\vec{r}=(+\hat{i}+3\hat{j}+5\hat{k})$ note that $\vec{r}.\vec{p}=0$ is parallel to :

(A) $\hat{i}+3\hat{j}-2\hat{k}$
(B) $3\hat{i}+\hat{j}+2\hat{k}$
(C) $-3\hat{i}-\hat{j}-2\hat{k}$
(D) $-\hat{i}+3\hat{j}+2\hat{k}$

Q16 JEE Main 2020 - 9 January (Evening)

An electric field $\vec{E}=4x\hat{i}-\left(y^2+1\right)\hat{j}\,N/C$ passes through the box shown in figure. The flux of the electric field through surface $ABCD$ and $BCGF$ and marked as ϕ_I and ϕ_{II} respectively. The



difference between $(\phi_I - \phi_{II})$ is $\left(\text{in } Nm^2/C\right)$

Answer Key

Q1 (B)	**Q2** (B)	**Q3** (D)	**Q4** (A)
Q5 (A)	**Q6** (C)	**Q7** (A)	**Q8** (C)
Q9 (A)	**Q10** (C)	**Q11** (C)	**Q12** (B)
Q13 (A)	**Q14** (C)	**Q15** (A)	**Q16** (48)

Q1

Let charges on shells be q_1 and q_2

$$\mathrm{q}_1+\mathrm{q}_2=\mathrm{Q} \quad \ldots \text{(i)}$$

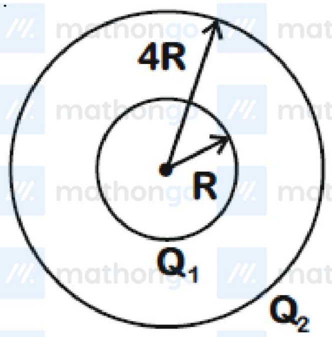
$$\frac{\mathrm{q}_1}{4\pi\mathrm{r}^2}=\frac{\mathrm{q}_2}{4\pi\mathrm{R}^2} \quad \ldots \text{(ii)}$$

We get $\mathrm{q}_1=\frac{\mathrm{r}^2}{\mathrm{r}^2+\mathrm{R}^2}\mathrm{Q}, \mathrm{q}_2=\frac{\mathrm{R}^2}{\mathrm{r}^2+\mathrm{R}^2}\mathrm{Q}$

$$\mathrm{V}=\frac{1}{4\pi\varepsilon_0}\left(\frac{\mathrm{q}_1}{\mathrm{r}}+\frac{\mathrm{q}_2}{\mathrm{R}}\right)$$

$$=\frac{1}{4\pi\varepsilon_0}\frac{(\mathrm{R}+\mathrm{r})}{\left(\mathrm{R}^2+\mathrm{r}^2\right)}\mathrm{Q}$$

Q2



$$\because \frac{Q_1}{R^2}=\frac{Q_2}{16R^2}$$

$$\Rightarrow Q_2=16Q_1$$

$$\left(\frac{KQ_1}{R}+\frac{KQ_2}{4R}\right)-\left(\frac{KQ_1}{4R}+\frac{KQ_2}{4R}\right)=V_1-V_2$$

$$\Rightarrow \quad V_1-V_2=\frac{3}{4}\times\frac{Q_1}{4\pi\varepsilon_0 R}=\frac{3Q_1}{16\pi\varepsilon_0 R}$$

Q3

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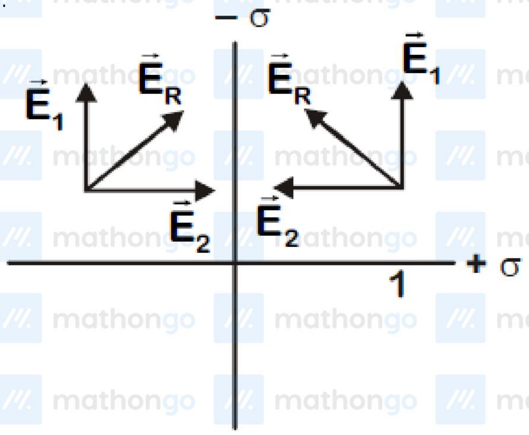

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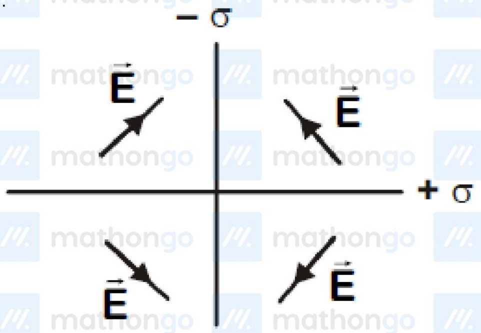


Q4

$$\Delta u = \frac{4q^2}{4\pi\varepsilon_0}\left[\frac{1}{3d/2} - \frac{1}{d/2}\right]$$

$$\Delta u = -\frac{4q^2}{3\pi\varepsilon_0 d}$$

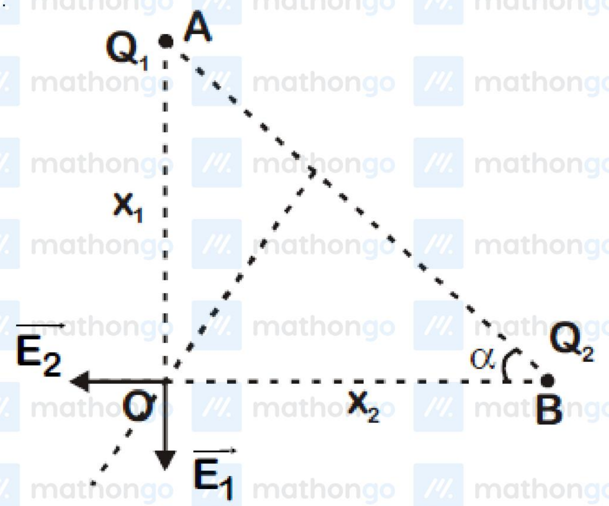
Q5

$$\Sigma Q = 0$$

$$\Rightarrow \quad V = 0$$

Also, field of one charge will get cancelled due to another symmetrical charge in front of it.

Q6



Net field along AB at O must be zero.

$\mathrm{E}_2 \cos \alpha=\mathrm{E}_1 \sin \alpha$

$\frac{kQ_2}{x_2^2} \cdot \frac{x_2}{AB}=\frac{kQ_1}{x_1^2} \cdot \frac{x_1}{AB}$

$\frac{Q_1}{Q_2}=\frac{x_1}{x_2}$

Q7

$\overrightarrow{\mathbf{p}}$ $\overrightarrow{\mathbf{p}}$

$\mathbf{U}=-\overrightarrow{\mathbf{p}} \cdot \overrightarrow{\mathbf{E}}$

$\overrightarrow{\mathrm{E}}=\frac{2 \overrightarrow{\mathrm{p}}}{4 \pi \varepsilon_0 \times \mathrm{a}^3}$

$\therefore \quad \Delta U_{\text{loss}}=\frac{2 p^2}{4 \pi \varepsilon_0 a^3}=2 \times \frac{1}{2} m v^2$

$\Rightarrow \quad \mathbf{v}=\frac{\mathbf{p}}{\mathbf{a}} \sqrt{\frac{1}{2 \pi \varepsilon_0 \mathbf{m a}}}$

Q8

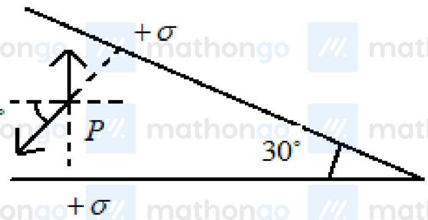
For spherical shell

$\mathrm{E}=\frac{1}{4 \pi \varepsilon_0} \frac{\mathrm{Q}}{\mathrm{r}^2} \quad(\mathrm{r}>\mathrm{R})$

$=0$

$(r < R)$
Force on charge in electric field
$\mathrm{F} = \mathrm{qE}$

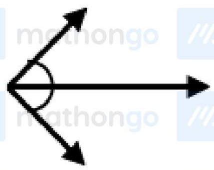
Q9



$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\cos 60^\circ\,(-\widehat{x}) + \left[\frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\sin 60^\circ\right](\hat{y})$$

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\left[\left(1 - \frac{\sqrt{3}}{2}\right)\hat{y} - \frac{1}{2}\widehat{x}\right]$$

Q10



$$E_{net} = \frac{4kq}{d^2} \times 2\cos 30^\circ = \frac{q\sqrt{3}}{\pi\varepsilon_0 d^2}$$

Q11

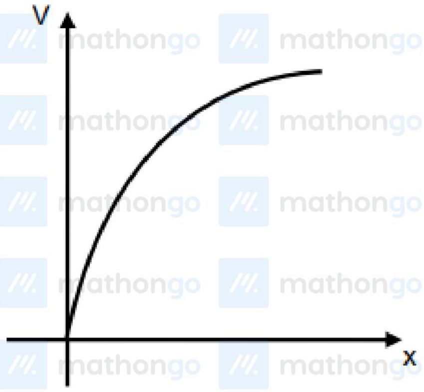
Magnitude of electric field is constant & the surface is equipotential

Q12

$$\frac{E_1}{E_2} = \frac{r_1}{r_2}$$

$$\frac{V_1}{V_2} = \frac{E_1 r_1}{E_2 r_2} = \frac{r_1}{r_2} \times \frac{r_1}{r_2} = \left(\frac{r_1}{r_2}\right)^2$$

Q13



$V^2 = \frac{2qE}{m}x$

Q14

For a solid sphere

$$E = \frac{\rho r}{3\varepsilon_0}$$

$$E_A = \frac{-\rho R}{2(3\varepsilon_0)}$$

$$|E_A| = \frac{\rho R}{6\varepsilon_0}$$

Electric field at point $B = E_B = E_{1A} + E_{2A}$

$E_{2A} =$ Electric Field Due to solid sphere of radius $R/2$ (which having charge density $-\rho$)

$$= -\frac{KQ' \times 4}{9R^2} = -\frac{\rho R}{54\varepsilon_0}$$

$$E_B = E_{1A} + E_{2A} = \frac{\rho R}{3\varepsilon_0} - \frac{\rho R}{54\varepsilon_0} = \frac{17\rho R}{54\varepsilon_0}$$

$$\frac{|E_A|}{|E_B|} = \frac{9}{17}$$

Q15

Since $\vec{p} \cdot \vec{r} = 0$

$\vec{E}$ must be anti-parallel to $\vec{p}$

So, $\vec{E} = -\lambda(\vec{p})$

where (λ) is a arbitrary positive constant

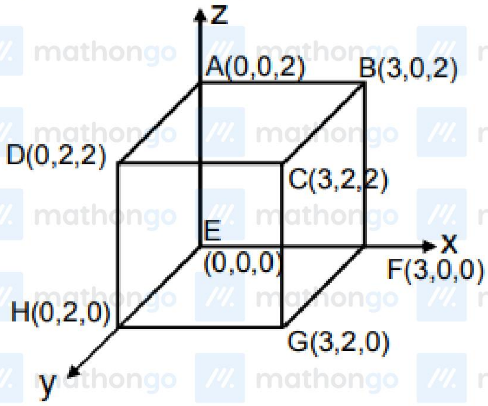
Now $\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$

$\vec{A} || \vec{E}$

$\frac{a}{\lambda} = \frac{b}{3\lambda} = \frac{c}{-2\lambda} = k$

so $\vec{A} = \lambda k\left(\hat{i} + 3\hat{j} - 2\hat{k}\right)$

Q16



Flux via $ABCD$

$\phi_1 = \int \vec{E} \cdot d\vec{A} = 0$

Flux via $BCEF$

$\phi_2 = \int \vec{E} \cdot d\vec{A}$

$\phi_2 = \vec{E} \cdot \vec{A} = \left(4x\hat{i} - \left(y^2 + 1\right)\hat{j}\right) \cdot 4\hat{i}$

$= 16x,\ x = 3$

$\phi_2 = 48\frac{N - m^2}{C}; \phi_1 - \phi_2 = -48\frac{N - m^2}{C}$